Frame Conditions & Neighbourhood Semantics

Chapter 2

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Frame conditions

Properties of the accessibility relation correspond to modal axioms.

Proposition 1

A frame $\mathfrak{F}_R$ validates **T** *($\Box A \to A$) if and only if R is reflexive.*

Likewise, transitivity corresponds to $\Box A \to \Box\Box A$ and symmetry to $A \to \Box\Diamond A$ [1, ch. 3].

Neighbourhood semantics

Generalise relational models by replacing R with a neighbourhood function [4, 3].

Definition 1

A *neighbourhood model* $\mathfrak{M}_N = \langle W, f_N, V \rangle$ has $f_N : W \to \wp(\wp(W))$, assigning each world a collection of neighbourhoods. The modal clause:

$$\mathfrak{M}_N, w \Vdash \Box A \quad \text{iff} \quad [\![A]\!]^{\mathfrak{M}_N} \in f_N(w).$$

Why bother?

Every Kripke frame induces a neighbourhood frame, but not conversely: neighbourhood frames model non-normal logics lacking **K** or Nec [2].

References I

[1] Patrick Blackburn, Maarten de Rijke, and Yde Venema. *Modal Logic*. Vol. 53. Cambridge Tracts in Theoretical Computer Science. Cambridge: Cambridge University Press, 2001. ISBN: 978-0521527149. DOI: 10.1017/CBO9781107050884.

[2] Brian F. Chellas. *Modal Logic: An Introduction*. Cambridge: Cambridge University Press, 1980. ISBN: 978-0-521-29515-4. DOI: 10.1017/CBO9780511621192.

[3] Eric Pacuit. *Neighborhood Semantics for Modal Logic*. Cham, Switzerland: Springer, 2017.

[4] Dana Scott. "Advice on Modal Logic". In: *Philosophical Problems in Logic*. Ed. by Karel Lambert. Vol. 29. Synthese Library. Dordrecht: Springer, 1970, pp. 143–173. DOI: 10.1007/978-94-010-3272-8_7.